In [6]:
```
import lightgbm as lgb
print(lgb.__version__)
```

```
4.7.0
```

In [ ]:
```
import pandas as pd
dataset=pd.read_csv("insurance.csv")
dataset
```

In [ ]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

In [ ]:
```
x=dataset[['age','bmi','children','sex_male','smoker_yes','region_northwest','region_southeast','region_southwest']]
y=dataset[['charges']]
print(x)
print(y)
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.20,random_state=42)
```

In [8]:
```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

In [12]:
```
from lightgbm import LGBMRegressor
reg=LGBMRegressor(
    objective='regression',
    metric='rmse',
    boosting_type='gbdt',
    num_leaves=31,
    learning_rate=0.05,
    feature_fraction=0.9
)
reg.fit(x_train,y_train)
```

```
[LightGBM] [Warning] feature_fraction is set=0.9, colsample_bytree=1.0 will be ignored. Current value: feature_fraction=0.9
[LightGBM] [Warning] feature_fraction is set=0.9, colsample_bytree=1.0 will be ignored. Current value: feature_fraction=0.9
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.000205 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 325
[LightGBM] [Info] Number of data points in the train set: 1070, number of used features: 8
[LightGBM] [Info] Start training from score 13346.089733
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[12]:

▾ LGBMRegressor

▸Parameters

▸Fitted attributes

In [ ]:
```
y_pred=reg.predict(x_test)
y_pred
```

In [14]:
```
from sklearn.metrics import r2_score
result=r2_score(y_pred,y_test)
result
```

Out[14]:

```
0.8497833819826975
```